Effect of mammalian lignans on the growth of prostate cancer cell lines.

BACKGROUND: Mammalian lignans, enterolactone (EL) and enterodiol (ED), have been shown to inhibit breast and colon carcinoma. To date, there have been no reports of the effect of lignans on prostatic carcinoma. We investigated the effects of ED and EL on three human prostate cancer cell lines (PC-3, DU-145 and LNCaP). MATERIALS AND METHODS: Cells were treated with either 0.1% (v/v) DMSO (vehicle) or 10-100 microM of EL, ED or genistein (positive control) for 72 hours. Cell viability was measured by the propidium iodide nuclei staining fluorometric assay with each assay performed in triplicate. RESULTS: At 10-100 microM, EL significantly inhibited the growth of all cell lines, whereas ED only inhibited PC-3 and LNCaP cells. While EL was a more potent growth inhibitor than ED, both were less potent than genistein. The dose for 50% growth inhibition of LNCaP cells (IC50) by EL was 57 microM, whereas IC50 was 100 microM for ED, (the observed IC50 for genistein was 25 microM). CONCLUSION: ED and EL suppress the growth of prostate cancer cells, and may do so via hormonally-dependent and independent mechanisms.